

TAMIL NADU STATE BOARD - STANDARD 10 MATHEMATICS

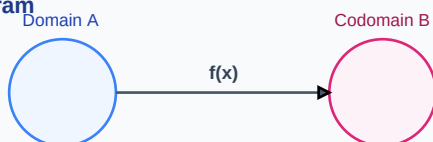
CHAPTER 8: STATISTICS AND PROBABILITY

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 10 English Medium
Subject:	Mathematics
Chapter Name:	Statistics and Probability

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram

Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Standard Deviation	The square root of the arithmetic mean of the squares of deviations of values from their arithmetic mean.
Coefficient of Variation (CV)	A relative measure of dispersion, expressed as percentage of standard deviation to mean.
Addition Theorem	For any two events A and B in a sample space, the probability of union is $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
Mutually Exclusive Events	Events that cannot occur simultaneously, meaning their intersection is an empty set with zero probability.

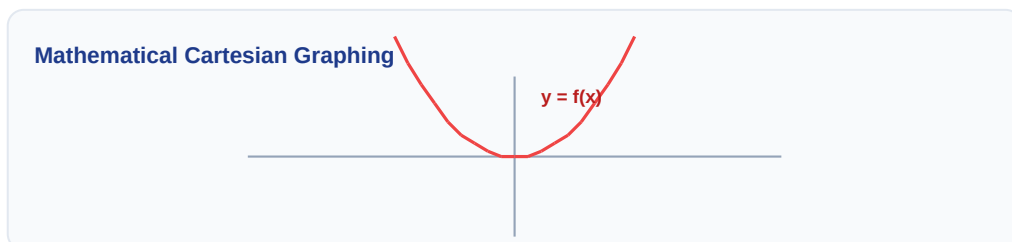
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Range = Largest value (L) - Smallest value (S)
- Standard Deviation (Ungrouped): $\sigma = \sqrt{\frac{\sum(x - \text{mean})^2}{n}}$
- Coefficient of Variation: $CV = \left(\frac{\sigma}{\text{mean}}\right) * 100 \%$
- Classical Probability: $P(E) = \frac{n(E)}{n(S)}$
- Sure Event Probability: $P(S) = 1$ | Impossible Event: $P(\text{empty}) = 0$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: A card is drawn from a pack of 52 cards. Find the probability of getting a red king.

Step-by-step Solution:

Total cards $n(S) = 52$. Red kings in a deck are two (King of Hearts and King of Diamonds).

$n(E) = 2$.

Probability $P(E) = n(E) / n(S) = 2 / 52 = 1 / 26$.

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: If three coins are tossed simultaneously, find the probability of getting at least two heads.

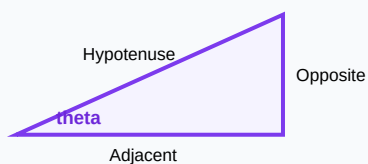
Analytical Solution Strategy:

Sample space $S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$. $n(S) = 8$.

Event E (at least 2 heads) = $\{HHH, HHT, HTH, THH\}$. $n(E) = 4$.

Probability $P(E) = n(E)/n(S) = 4/8 = 0.5$.

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Find the range of the data set: 25, 67, 48, 53, 18, 39, 44.

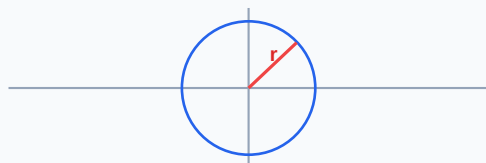
Q2 (2-Mark): If the variance of a data set is 64, find its standard deviation.

Q3 (2-Mark): If $P(A) = 0.5$, $P(B) = 0.6$, and $P(A \cap B) = 0.2$, find $P(A \cup B)$.

Q4 (5-Mark): Find the probability of getting a prime number when a fair die is rolled.

Q5 (5-Mark): The mean of a data is 25 and its standard deviation is 5. Find the coefficient of variation.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Stock Market Analytics

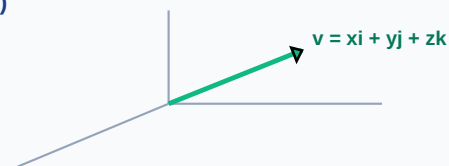
Standard deviation measures asset volatility, helping investors evaluate risk models in stock portfolios.

Application Case: Insurance Risk Modeling

Actuarial departments use probability distributions to calculate life expectancy and premium policy costs.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

